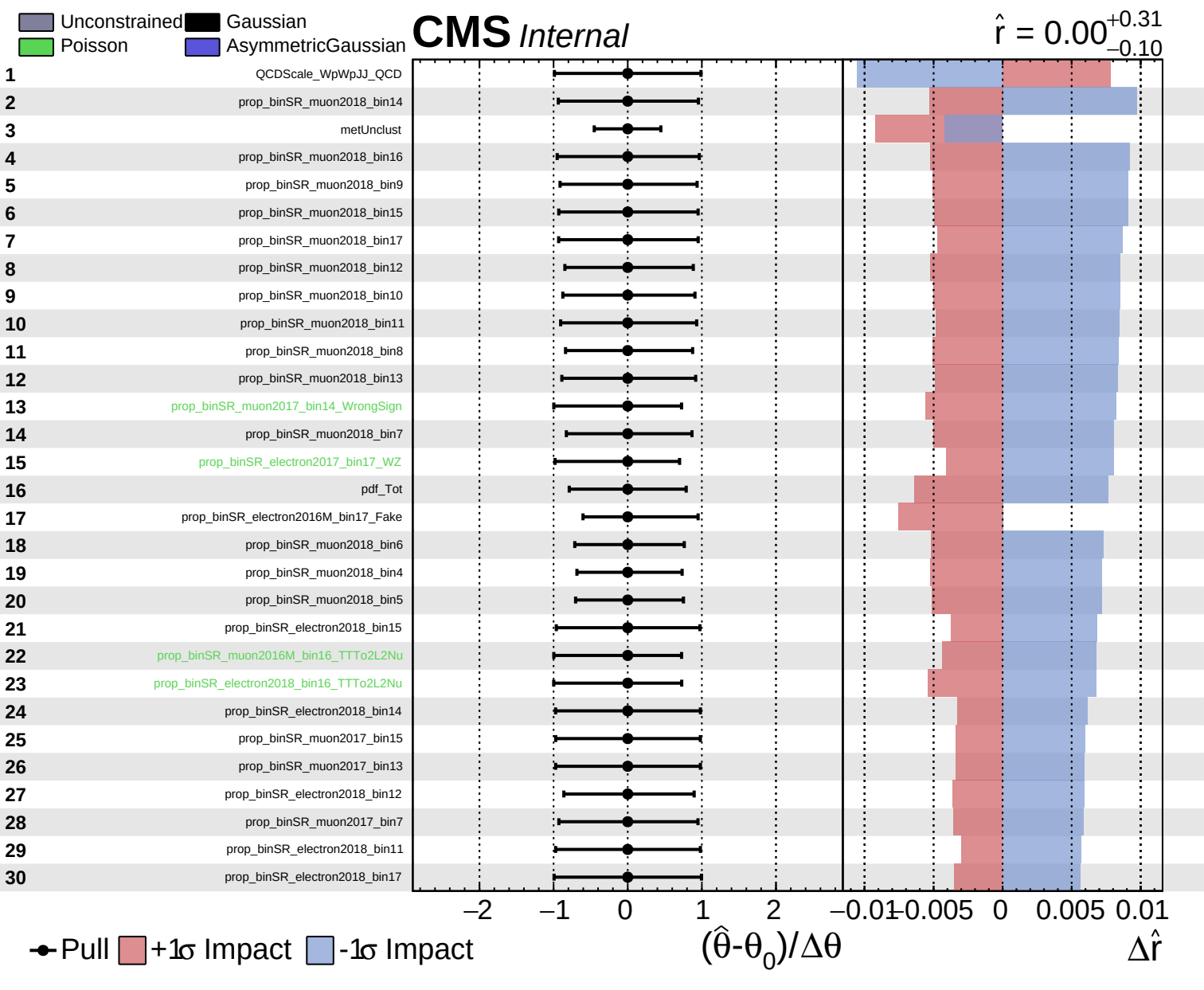
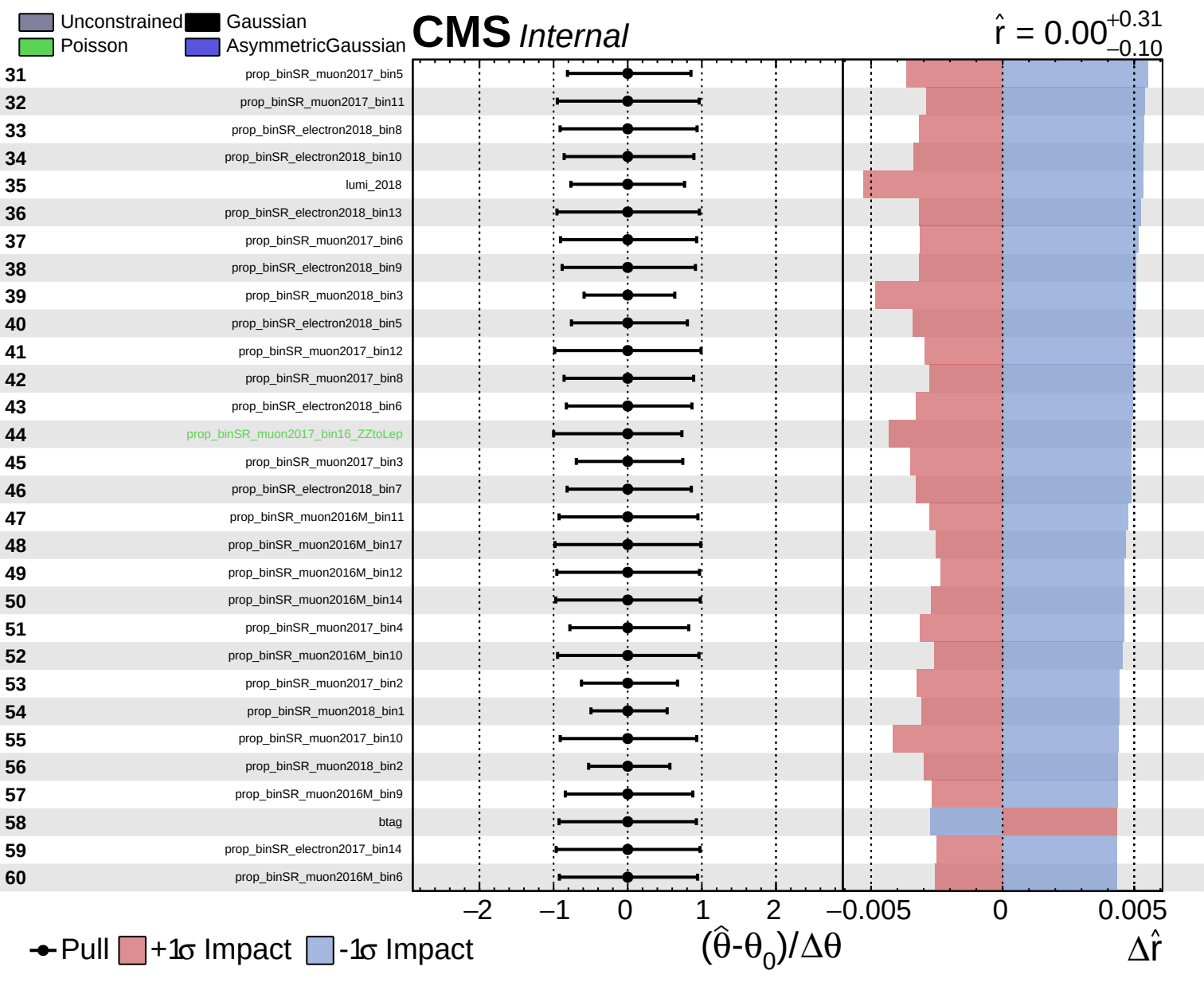
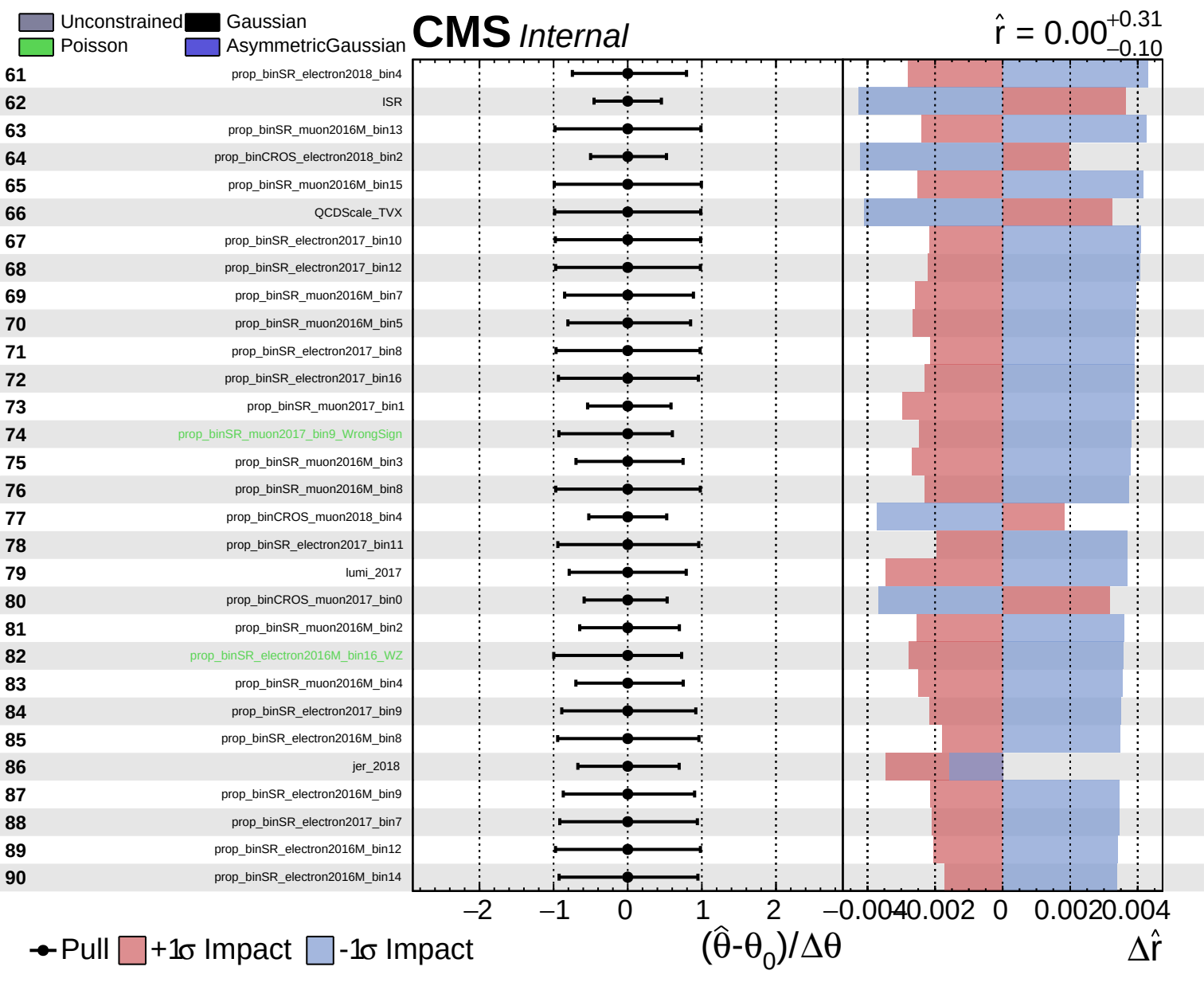
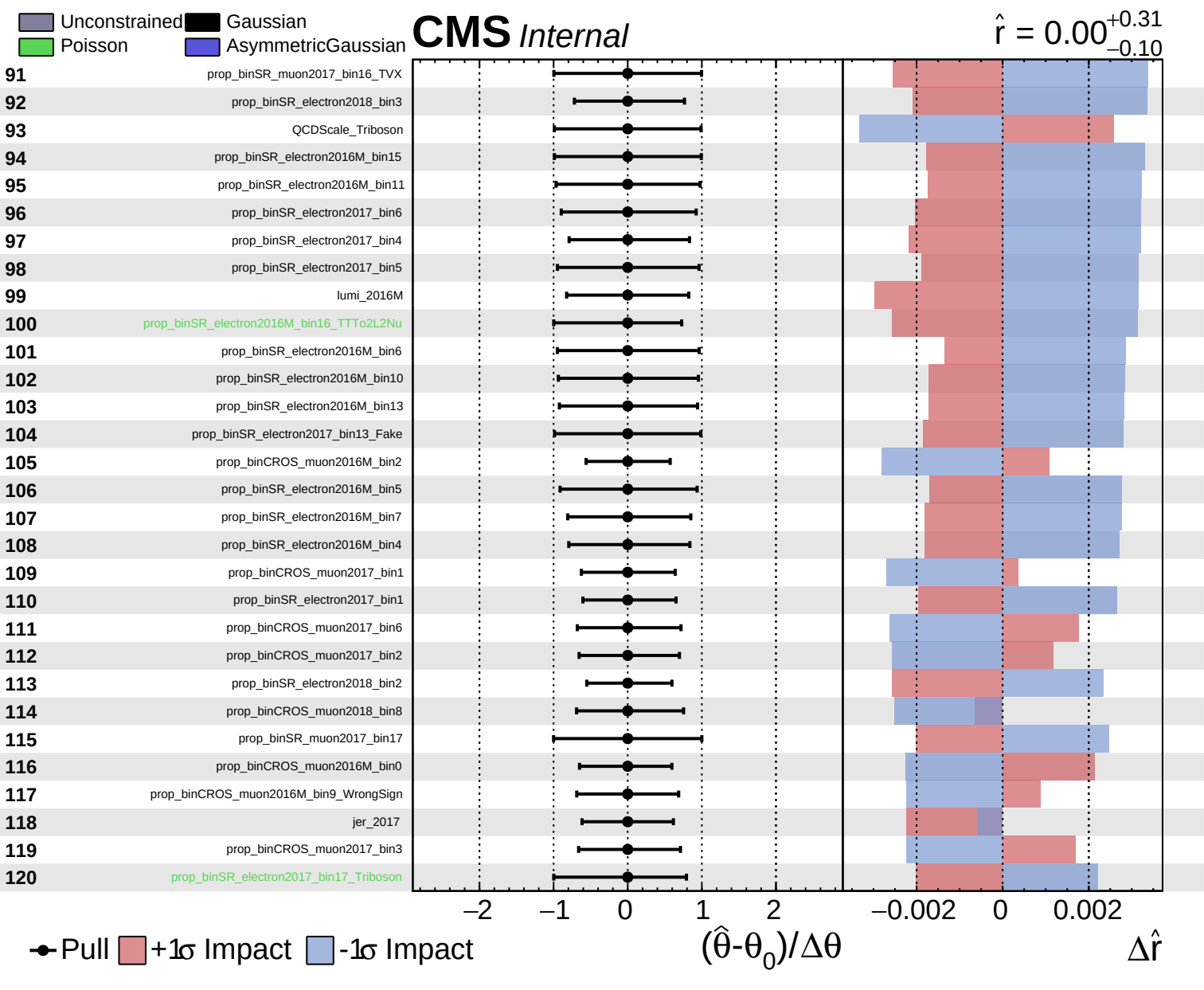
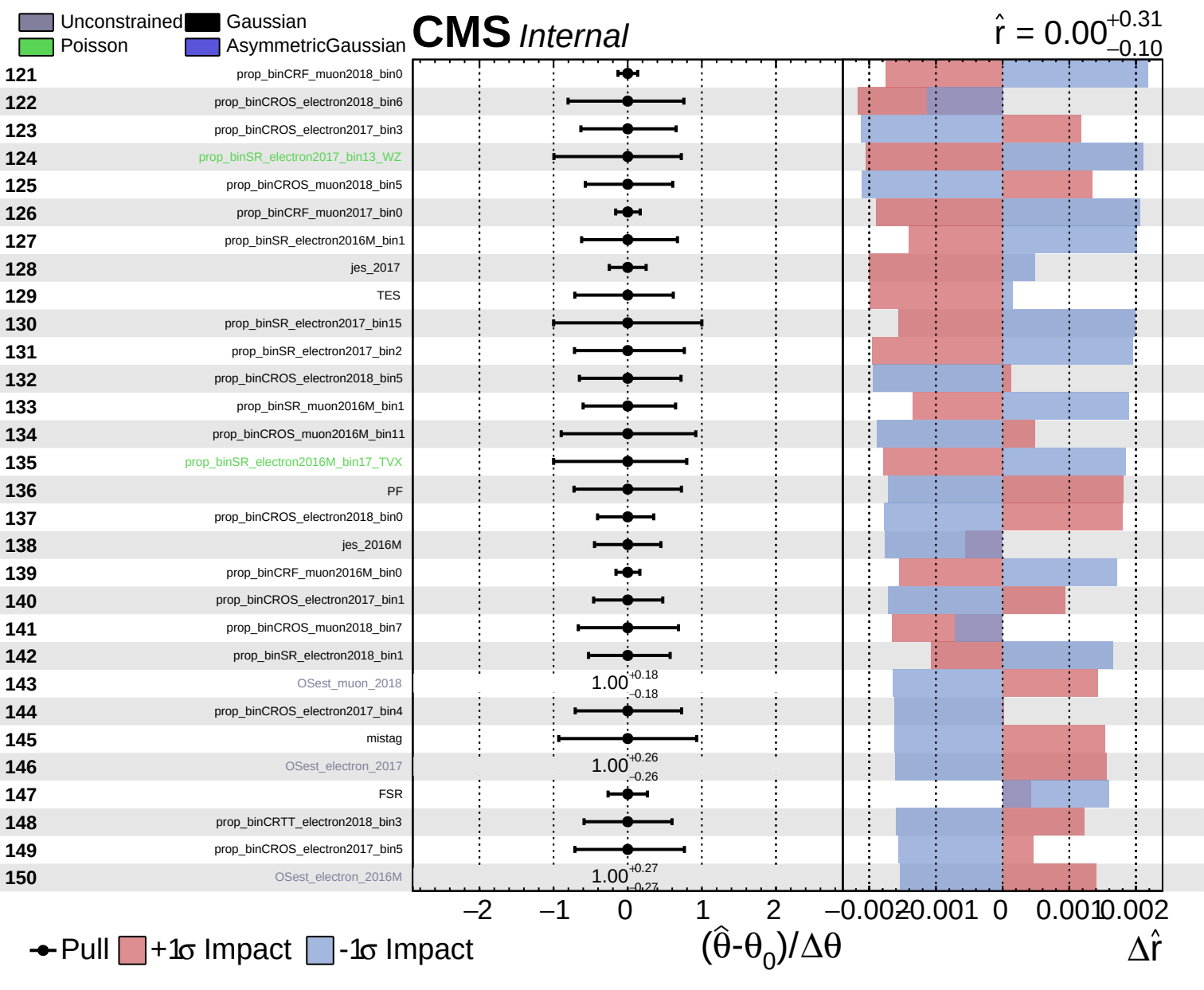
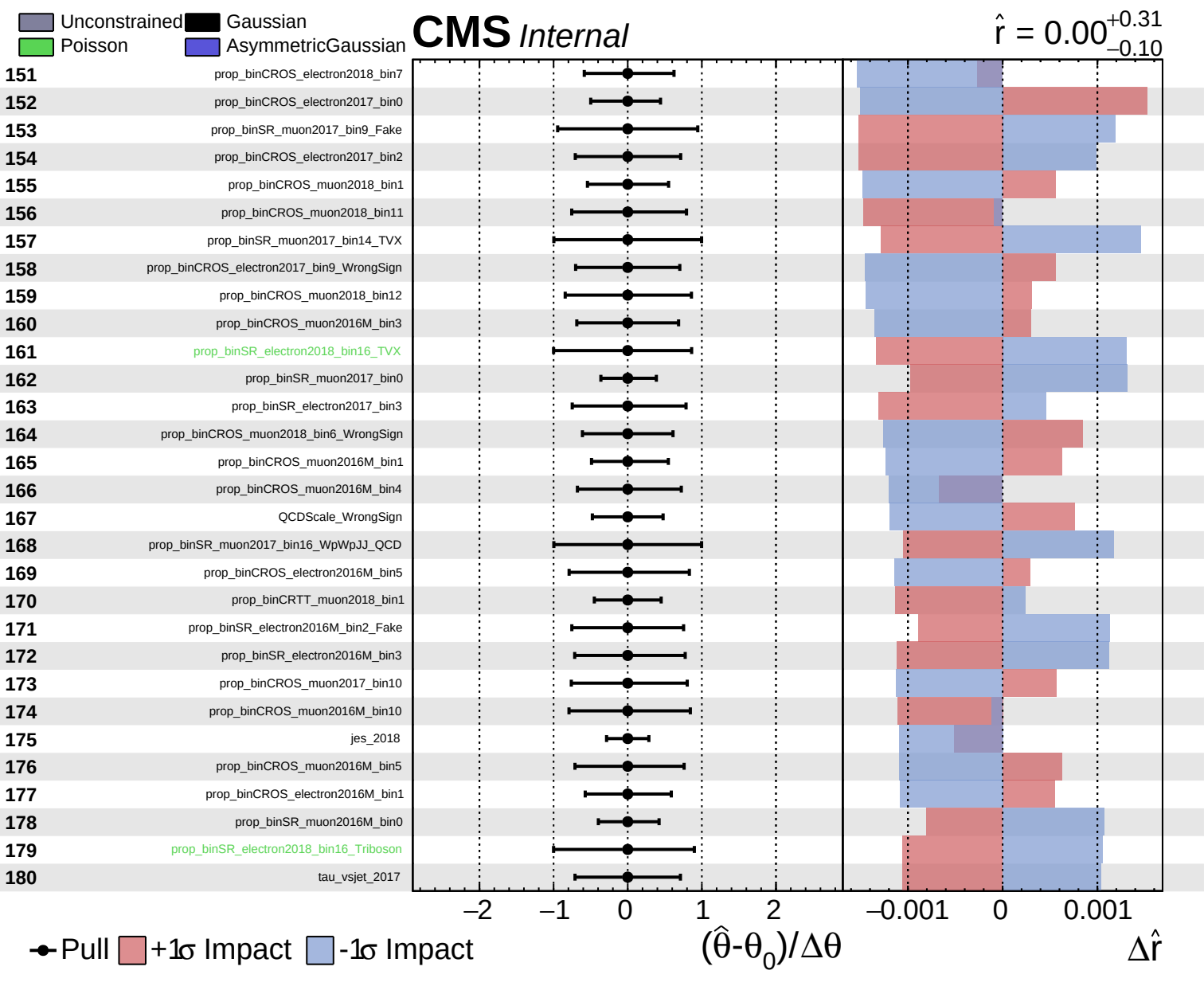

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
1 QCDScale_WpWpJJ_QCD
2 prop_binSR_muon2018_bin14
3 metUnclust
4 prop_binSR_muon2018_bin16
5 prop_binSR_muon2018_bin9
6 prop_binSR_muon2018_bin15
7 prop_binSR_muon2018_bin17
8 prop_binSR_muon2018_bin12
9 prop_binSR_muon2018_bin10
10 prop_binSR_muon2018_bin11
11 prop_binSR_muon2018_bin8
12 prop_binSR_muon2018_bin13
13 prop_binSR_muon2017_bin14_WrongSign
14 prop_binSR_muon2018_bin7
15 prop_binSR_electron2017_bin17_WZ
16 pdf_Tot
17 prop_binSR_electron2016M_bin17_Fake
18 prop_binSR_muon2018_bin6
19 prop_binSR_muon2018_bin4
20 prop_binSR_muon2018_bin5
21 prop_binSR_electron2018_bin15
22 prop_binSR_muon2016M_bin16_TTTo2L2Nu
23 prop_binSR_electron2018_bin16_TTTo2L2Nu
24 prop_binSR_electron2018_bin14
25 prop_binSR_muon2017_bin15
26 prop_binSR_muon2017_bin13
27 prop_binSR_electron2018_bin12
28 prop_binSR_muon2017_bin7
29 prop_binSR_electron2018_bin11
30 prop_binSR_electron2018_bin17
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

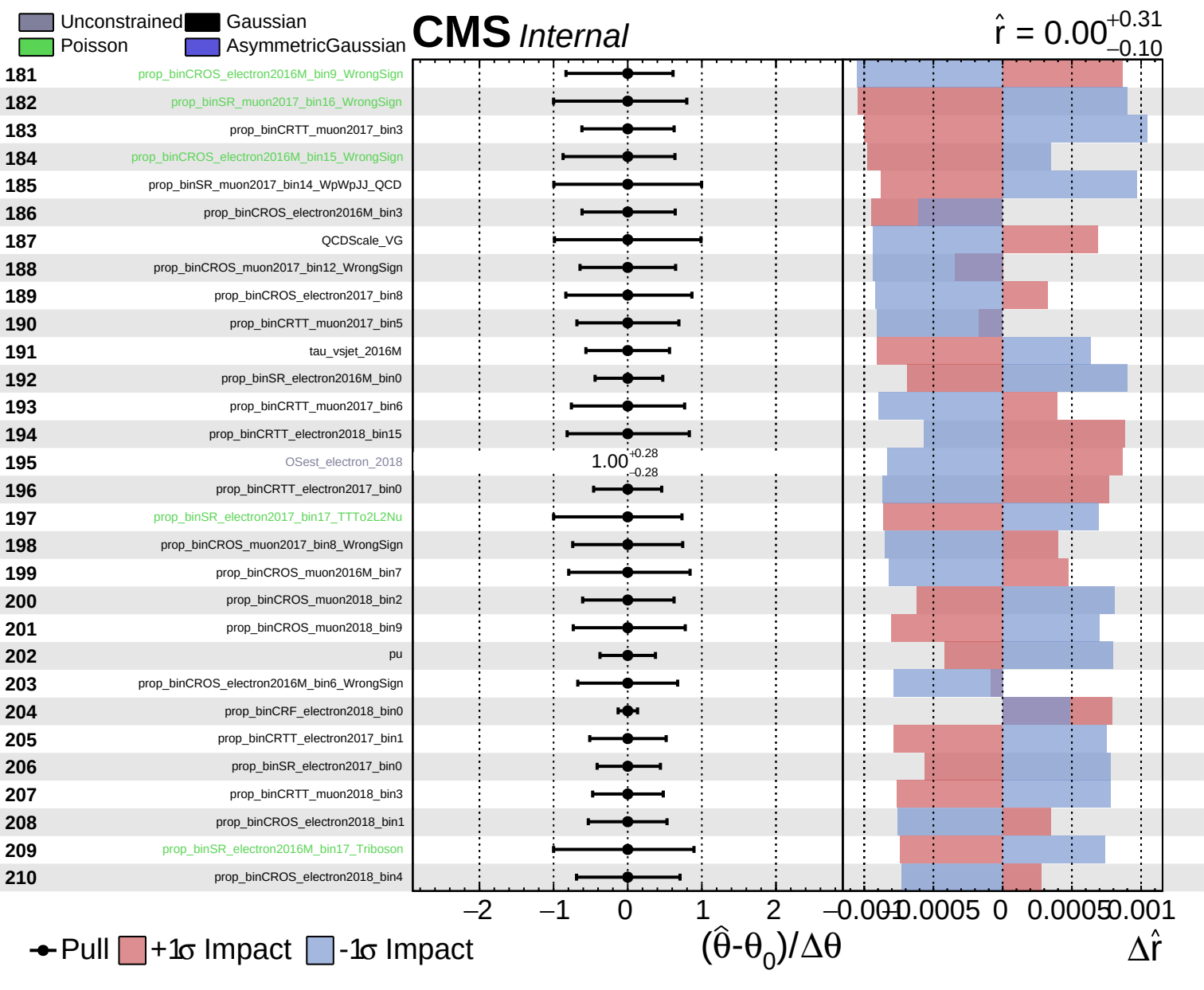

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
31 prop_binSR_muon2017_bin5
32 prop_binSR_muon2017_bin11
33 prop_binSR_electron2018_bin8
34 prop_binSR_electron2018_bin10
35 lumi_2018
36 prop_binSR_electron2018_bin13
37 prop_binSR_muon2017_bin6
38 prop_binSR_electron2018_bin9
39 prop_binSR_muon2018_bin3
40 prop_binSR_electron2018_bin5
41 prop_binSR_muon2017_bin12
42 prop_binSR_muon2017_bin8
43 prop_binSR_electron2018_bin6
44 prop_binSR_muon2017_bin16_ZZtoLep
45 prop_binSR_muon2017_bin3
46 prop_binSR_electron2018_bin7
47 prop_binSR_muon2016M_bin11
48 prop_binSR_muon2016M_bin17
49 prop_binSR_muon2016M_bin12
50 prop_binSR_muon2016M_bin14
51 prop_binSR_muon2017_bin4
52 prop_binSR_muon2016M_bin10
53 prop_binSR_muon2017_bin2
54 prop_binSR_muon2018_bin1
55 prop_binSR_muon2017_bin10
56 prop_binSR_muon2018_bin2
57 prop_binSR_muon2016M_bin9
58 btag
59 prop_binSR_electron2017_bin14
60 prop_binSR_muon2016M_bin6
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

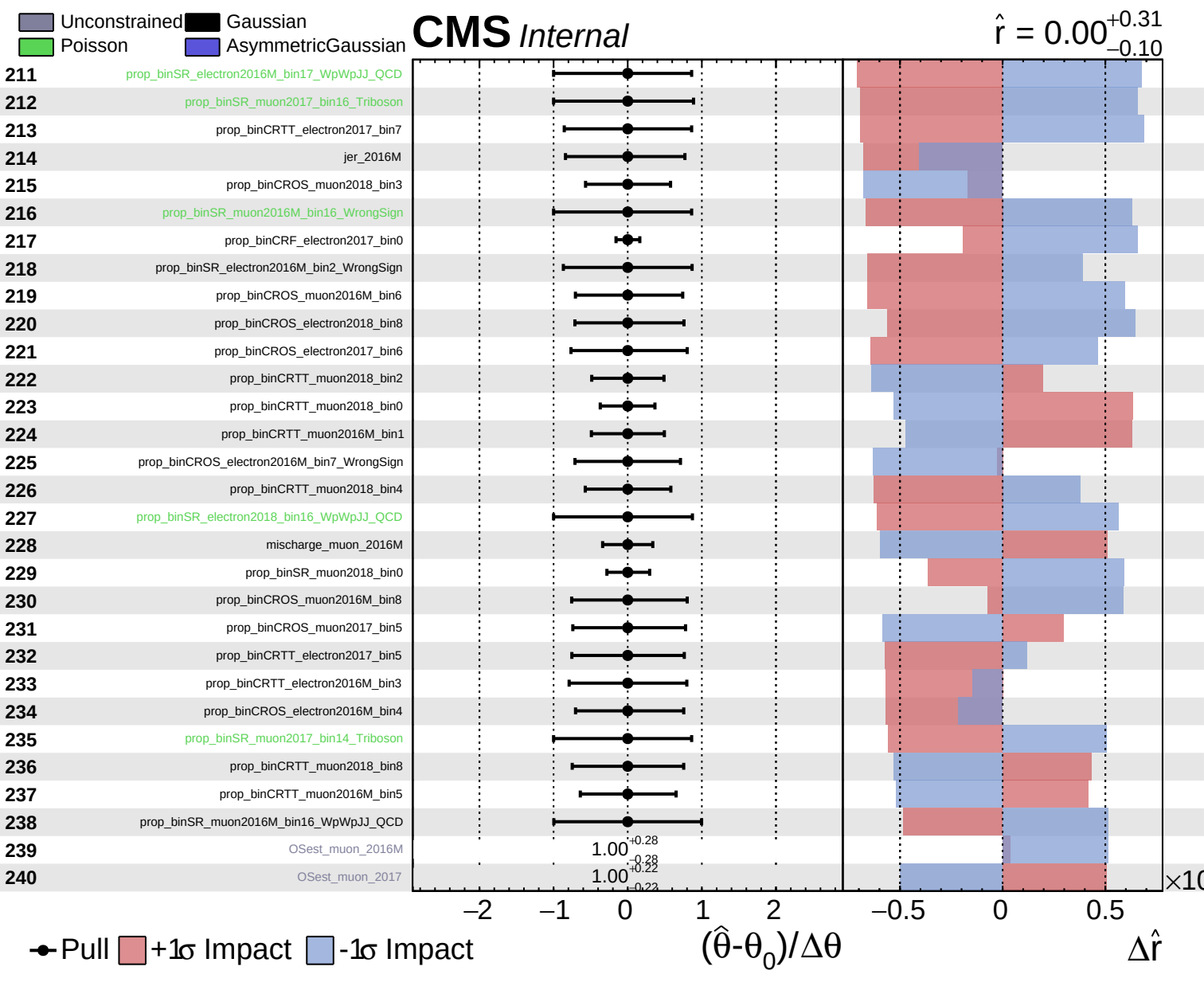

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
61 prop_binSR_electron2018_bin4
62 ISR
63 prop_binSR_muon2016M_bin13
64 prop_binCROS_electron2018_bin2
65 prop_binSR_muon2016M_bin15
66 QCDScale_TVX
67 prop_binSR_electron2017_bin10
68 prop_binSR_electron2017_bin12
69 prop_binSR_muon2016M_bin7
70 prop_binSR_muon2016M_bin5
71 prop_binSR_electron2017_bin8
72 prop_binSR_electron2017_bin16
73 prop_binSR_muon2017_bin1
74 prop_binSR_muon2017_bin9_WrongSign
75 prop_binSR_muon2016M_bin3
76 prop_binSR_muon2016M_bin8
77 prop_binCROS_muon2018_bin4
78 prop_binSR_electron2017_bin11
79 lumi_2017
80 prop_binCROS_muon2017_bin0
81 prop_binSR_muon2016M_bin2
82 prop_binSR_electron2016M_bin16_WZ
83 prop_binSR_muon2016M_bin4
84 prop_binSR_electron2017_bin9
85 prop_binSR_electron2016M_bin8
86 jer_2018
87 prop_binSR_electron2016M_bin9
88 prop_binSR_electron2017_bin7
89 prop_binSR_electron2016M_bin12
90 prop_binSR_electron2016M_bin14
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

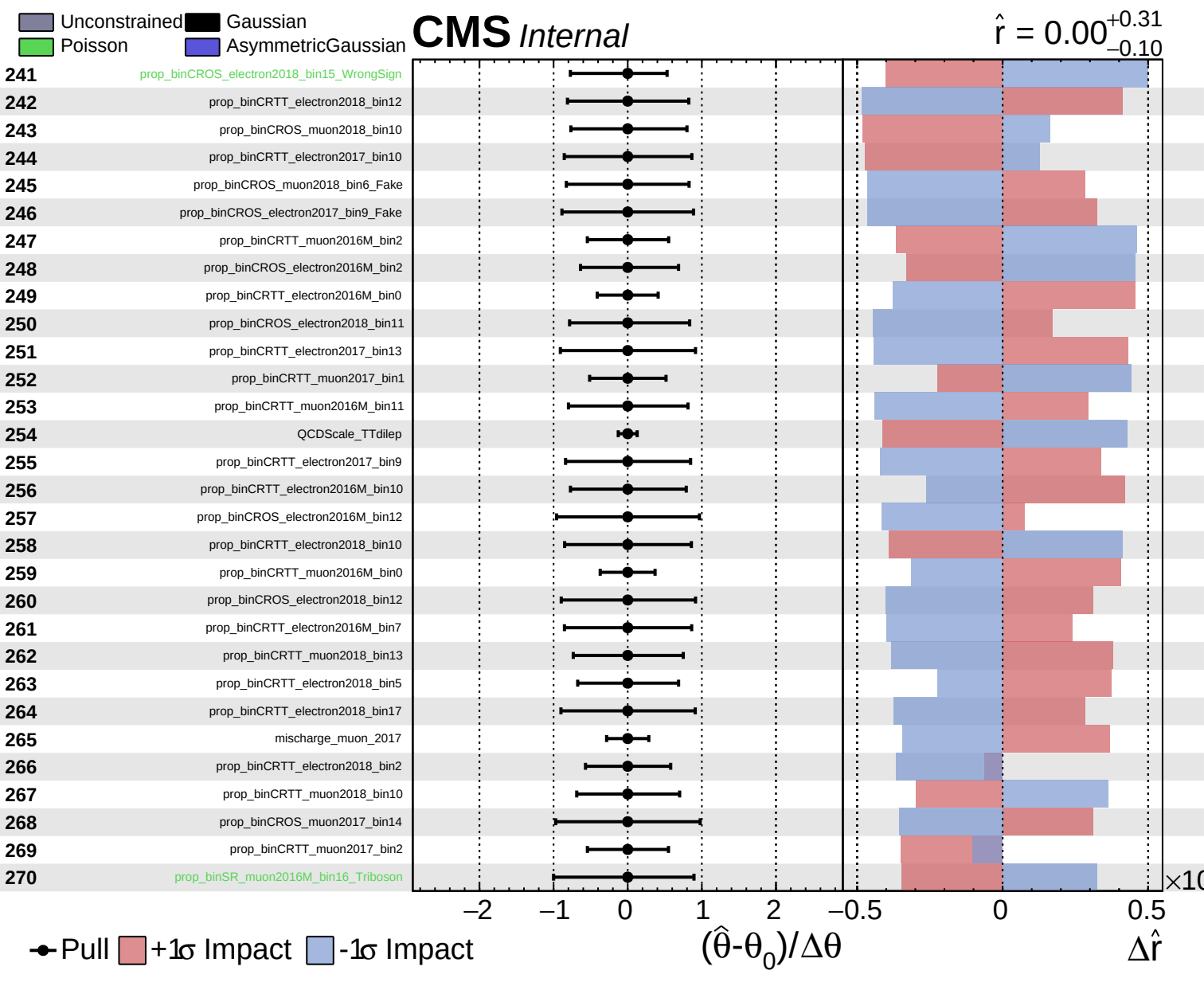

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00+0.31−0.10
91 prop_binSR_muon2017_bin16_TVX
92 prop_binSR_electron2018_bin3
93 QCDScale_Triboson
94 prop_binSR_electron2016M_bin15
95 prop_binSR_electron2016M_bin11
96 prop_binSR_electron2017_bin6
97 prop_binSR_electron2017_bin4
98 prop_binSR_electron2017_bin5
99 lumi_2016M
100 prop_binSR_electron2016M_bin16_TTTo2L2Nu
101 prop_binSR_electron2016M_bin6
102 prop_binSR_electron2016M_bin10
103 prop_binSR_electron2016M_bin13
104 prop_binSR_electron2017_bin13_Fake
105 prop_binCROS_muon2016M_bin2
106 prop_binSR_electron2016M_bin5
107 prop_binSR_electron2016M_bin7
108 prop_binSR_electron2016M_bin4
109 prop_binCROS_muon2017_bin1
110 prop_binSR_electron2017_bin1
111 prop_binCROS_muon2017_bin6
112 prop_binCROS_muon2017_bin2
113 prop_binSR_electron2018_bin2
114 prop_binCROS_muon2018_bin8
115 prop_binSR_muon2017_bin17
116 prop_binCROS_muon2016M_bin0
117 prop_binCROS_muon2016M_bin9_WrongSign
118 jer_2017
119 prop_binCROS_muon2017_bin3
120 prop_binSR_electron2017_bin17_Triboson
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

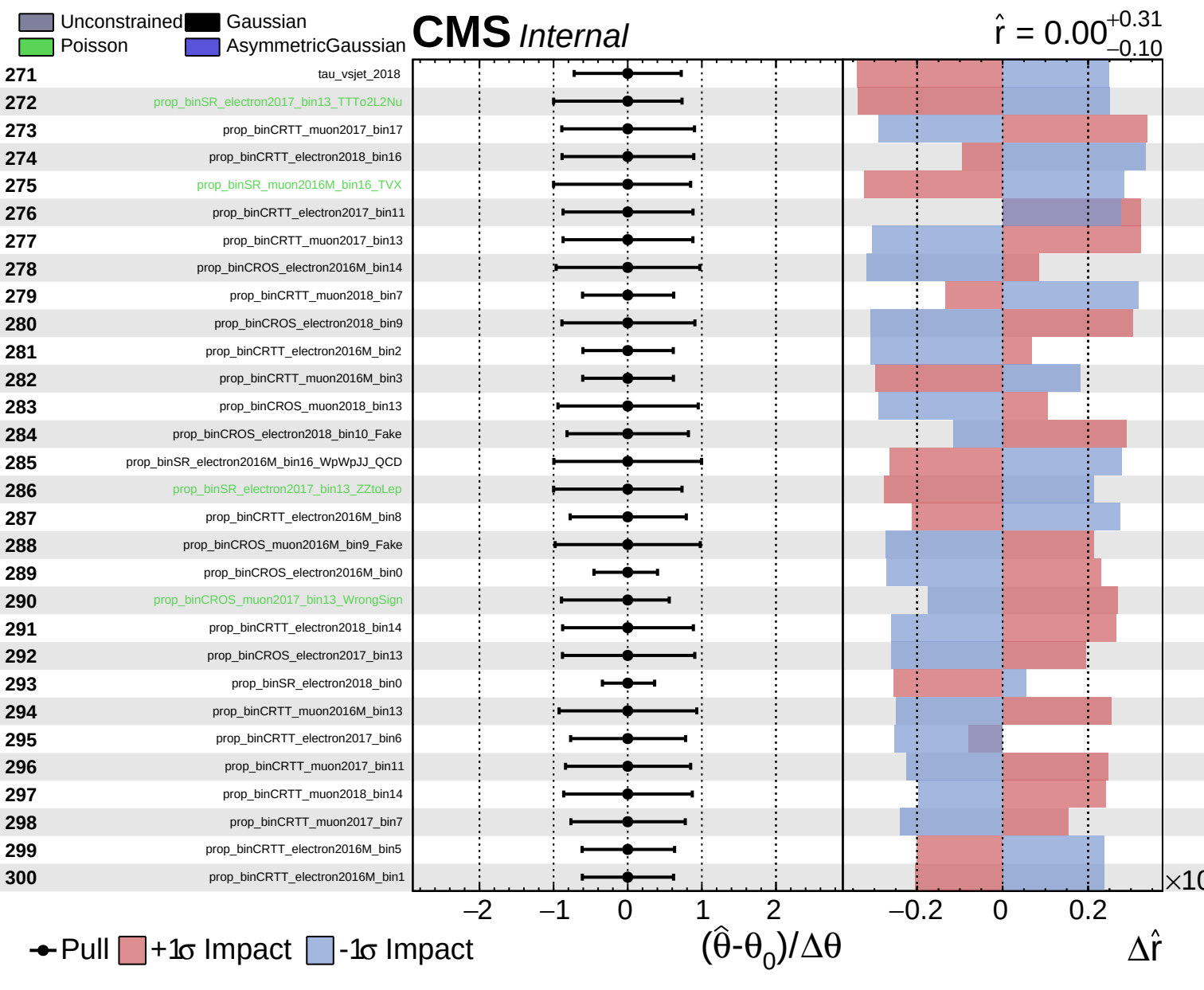

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
121 prop_binCRF_muon2018_bin0
122 prop_binCROS_electron2018_bin6
123 prop_binCROS_electron2017_bin3
124 prop_binSR_electron2017_bin13_WZ
125 prop_binCROS_muon2018_bin5
126 prop_binCRF_muon2017_bin0
127 prop_binSR_electron2016M_bin1
128 jes_2017
129 TES
130 prop_binSR_electron2017_bin15
131 prop_binSR_electron2017_bin2
132 prop_binCROS_electron2018_bin5
133 prop_binSR_muon2016M_bin1
134 prop_binCROS_muon2016M_bin11
135 prop_binSR_electron2016M_bin17_TVX
136 PF
137 prop_binCROS_electron2018_bin0
138 jes_2016M
139 prop_binCRF_muon2016M_bin0
140 prop_binCROS_electron2017_bin1
141 prop_binCROS_muon2018_bin7
142 prop_binSR_electron2018_bin1
143 OSest_muon_2018
1.00 +0.18 −0.18
144 prop_binCROS_electron2017_bin4
145 mistag
146 OSest_electron_2017
1.00 +0.26 −0.26
147 FSR
148 prop_binCRTT_electron2018_bin3
149 prop_binCROS_electron2017_bin5
150 OSest_electron_2016M
1.00 +0.27 −0.27
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

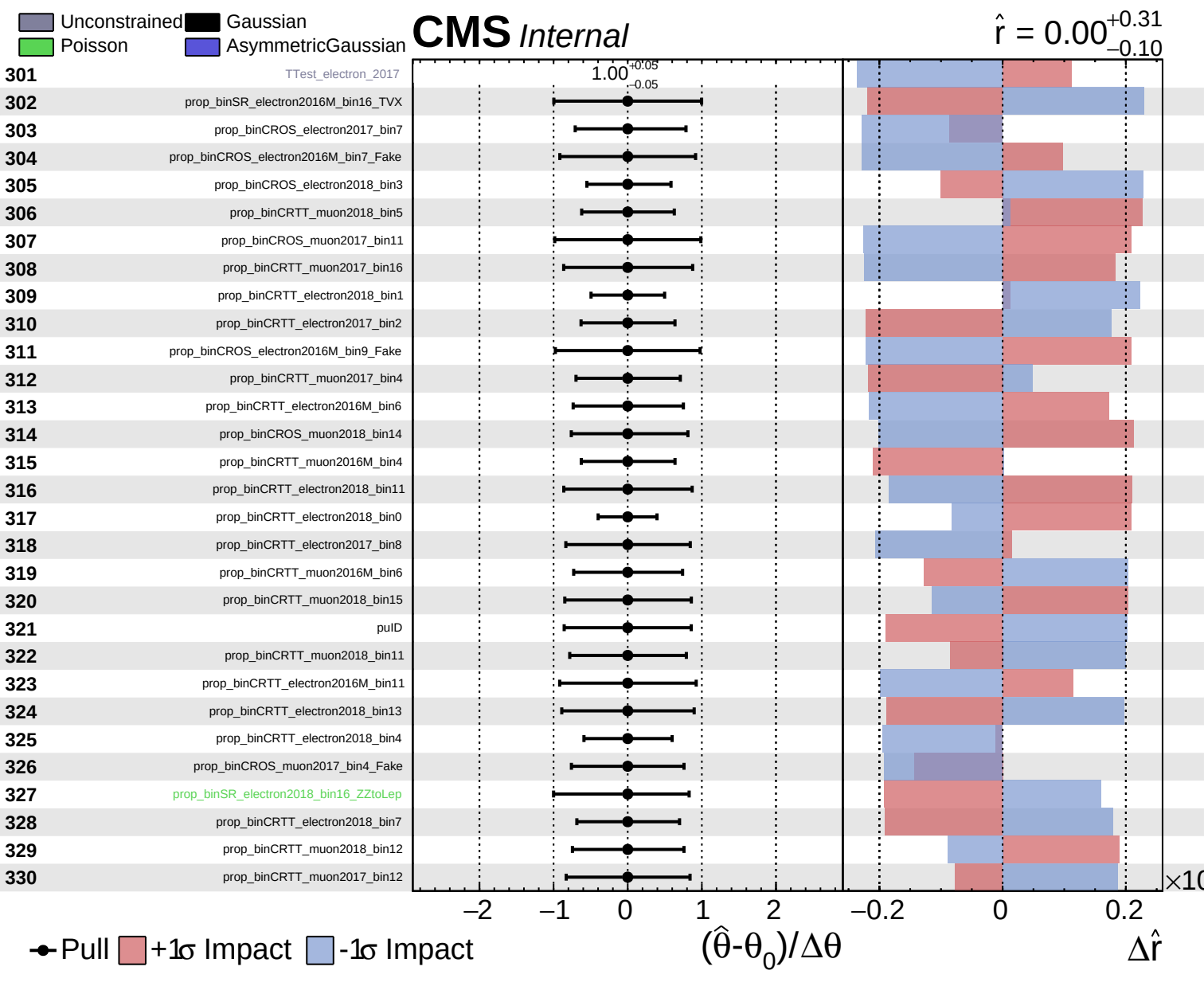

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
151 prop_binCROS_electron2018_bin7
152 prop_binCROS_electron2017_bin0
153 prop_binSR_muon2017_bin9_Fake
154 prop_binCROS_electron2017_bin2
155 prop_binCROS_muon2018_bin1
156 prop_binCROS_muon2018_bin11
157 prop_binSR_muon2017_bin14_TVX
158 prop_binCROS_electron2017_bin9_WrongSign
159 prop_binCROS_muon2018_bin12
160 prop_binCROS_muon2016M_bin3
161 prop_binSR_electron2018_bin16_TVX
162 prop_binSR_muon2017_bin0
163 prop_binSR_electron2017_bin3
164 prop_binCROS_muon2018_bin6_WrongSign
165 prop_binCROS_muon2016M_bin1
166 prop_binCROS_muon2016M_bin4
167 QCDScale_WrongSign
168 prop_binSR_muon2017_bin16_WpWpJJ_QCD
169 prop_binCROS_electron2016M_bin5
170 prop_binCRTT_muon2018_bin1
171 prop_binSR_electron2016M_bin2_Fake
172 prop_binSR_electron2016M_bin3
173 prop_binCROS_muon2017_bin10
174 prop_binCROS_muon2016M_bin10
175 jes_2018
176 prop_binCROS_muon2016M_bin5
177 prop_binCROS_electron2016M_bin1
178 prop_binSR_muon2016M_bin0
179 prop_binSR_electron2018_bin16_Triboson
180 tau_vsjet_2017
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

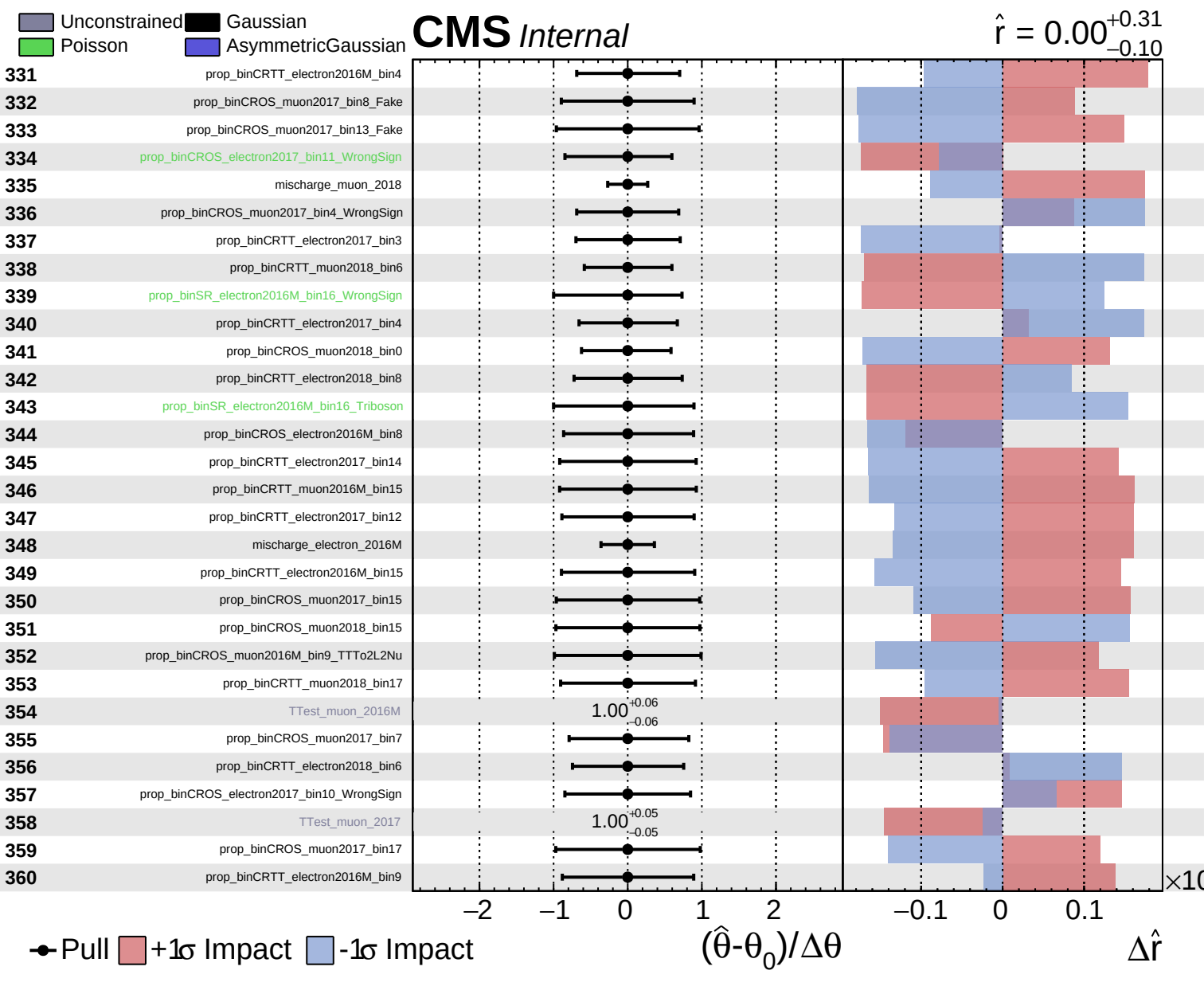

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
181 prop_binCROS_electron2016M_bin9_WrongSign
182 prop_binSR_muon2017_bin16_WrongSign
183 prop_binCRTT_muon2017_bin3
184 prop_binCROS_electron2016M_bin15_WrongSign
185 prop_binSR_muon2017_bin14_WpWpJJ_QCD
186 prop_binCROS_electron2016M_bin3
187 QCDScale_VG
188 prop_binCROS_muon2017_bin12_WrongSign
189 prop_binCROS_electron2017_bin8
190 prop_binCRTT_muon2017_bin5
191 tau_vsjet_2016M
192 prop_binSR_electron2016M_bin0
193 prop_binCRTT_muon2017_bin6
194 prop_binCRTT_electron2018_bin15
195 OSest_electron_2018
1.00 +0.28 -0.28
196 prop_binCRTT_electron2017_bin0
197 prop_binSR_electron2017_bin17_TTTo2L2Nu
198 prop_binCROS_muon2017_bin8_WrongSign
199 prop_binCROS_muon2016M_bin7
200 prop_binCROS_muon2018_bin2
201 prop_binCROS_muon2018_bin9
202 pu
203 prop_binCROS_electron2016M_bin6_WrongSign
204 prop_binCRF_electron2018_bin0
205 prop_binCRTT_electron2017_bin1
206 prop_binSR_electron2017_bin0
207 prop_binCRTT_muon2018_bin3
208 prop_binCROS_electron2018_bin1
209 prop_binSR_electron2016M_bin17_Triboson
210 prop_binCROS_electron2018_bin4
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

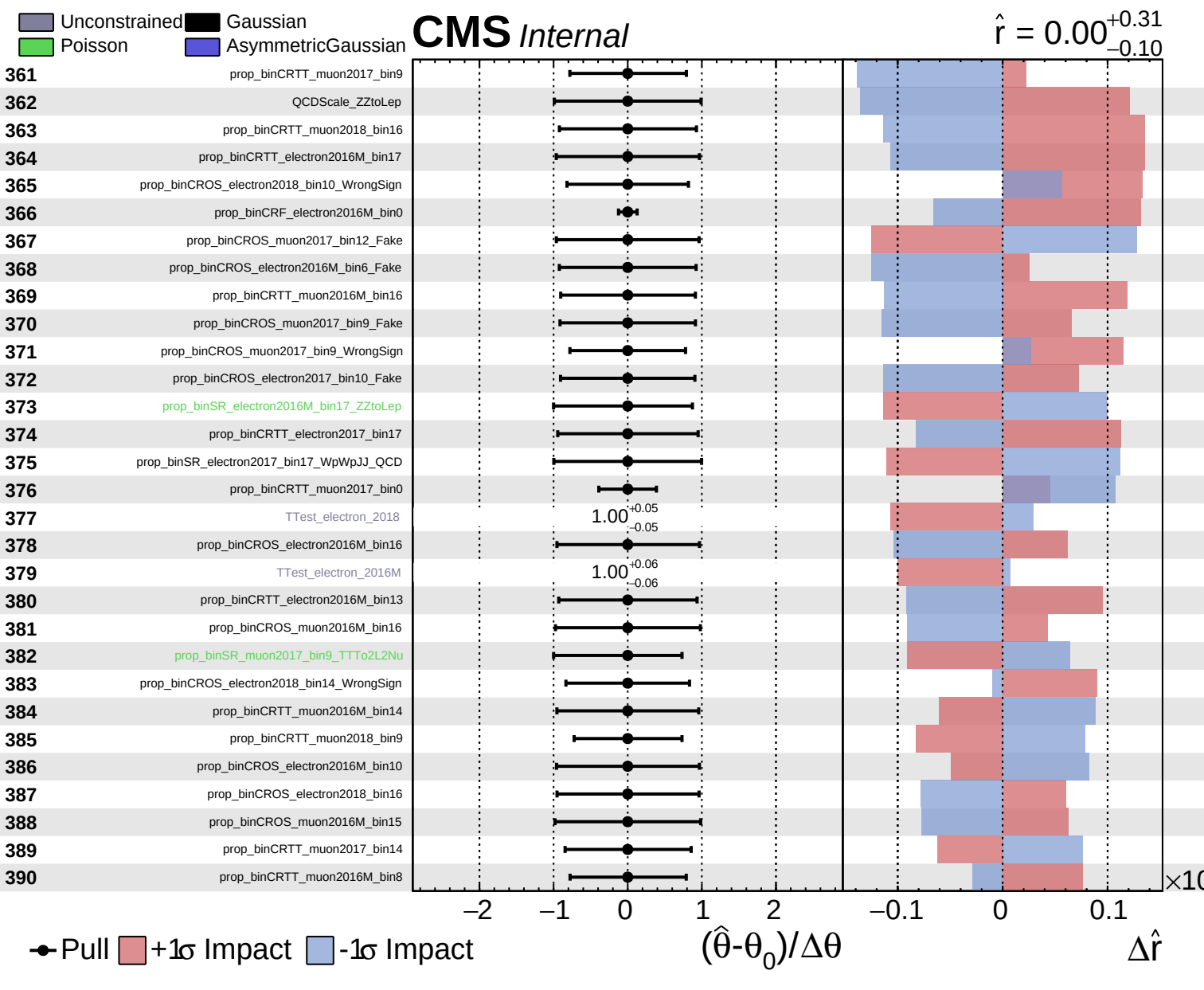

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
211 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
212 prop_binSR_muon2017_bin16_Triboson
213 prop_binCRTT_electron2017_bin7
214 jer_2016M
215 prop_binCROS_muon2018_bin3
216 prop_binSR_muon2016M_bin16_WrongSign
217 prop_binCRF_electron2017_bin0
218 prop_binSR_electron2016M_bin2_WrongSign
219 prop_binCROS_muon2016M_bin6
220 prop_binCROS_electron2018_bin8
221 prop_binCROS_electron2017_bin6
222 prop_binCRTT_muon2018_bin2
223 prop_binCRTT_muon2018_bin0
224 prop_binCRTT_muon2016M_bin1
225 prop_binCROS_electron2016M_bin7_WrongSign
226 prop_binCRTT_muon2018_bin4
227 prop_binSR_electron2018_bin16_WpWpJJ_QCD
228 mischarge_muon_2016M
229 prop_binSR_muon2018_bin0
230 prop_binCROS_muon2016M_bin8
231 prop_binCROS_muon2017_bin5
232 prop_binCRTT_electron2017_bin5
233 prop_binCRTT_electron2016M_bin3
234 prop_binCROS_electron2016M_bin4
235 prop_binSR_muon2017_bin14_Triboson
236 prop_binCRTT_muon2018_bin8
237 prop_binCRTT_muon2016M_bin5
238 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
239 OSest_muon_2016M
240 OSest_muon_2017
1.00 +0.28 −0.28
1.00 +0.22 −0.22
×10
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

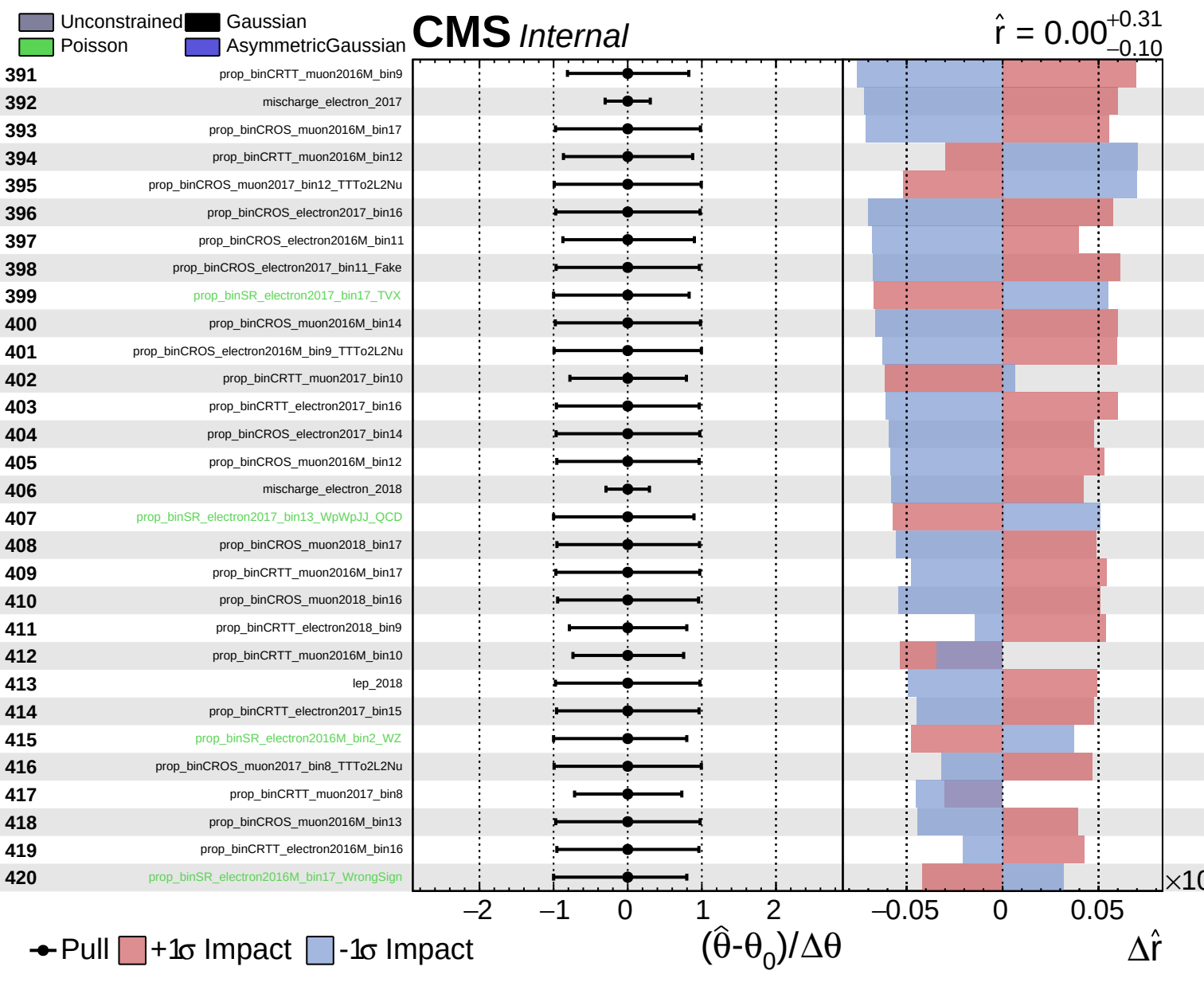

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
241 prop_binCROS_electron2018_bin15_WrongSign
242 prop_binCRTT_electron2018_bin12
243 prop_binCROS_muon2018_bin10
244 prop_binCRTT_electron2017_bin10
245 prop_binCROS_muon2018_bin6_Fake
246 prop_binCROS_electron2017_bin9_Fake
247 prop_binCRTT_muon2016M_bin2
248 prop_binCROS_electron2016M_bin2
249 prop_binCRTT_electron2016M_bin0
250 prop_binCROS_electron2018_bin11
251 prop_binCRTT_electron2017_bin13
252 prop_binCRTT_muon2017_bin1
253 prop_binCRTT_muon2016M_bin11
254 QCDScale_TTdilep
255 prop_binCRTT_electron2017_bin9
256 prop_binCRTT_electron2016M_bin10
257 prop_binCROS_electron2016M_bin12
258 prop_binCRTT_electron2018_bin10
259 prop_binCRTT_muon2016M_bin0
260 prop_binCROS_electron2018_bin12
261 prop_binCRTT_electron2016M_bin7
262 prop_binCRTT_muon2018_bin13
263 prop_binCRTT_electron2018_bin5
264 prop_binCRTT_electron2018_bin17
265 mischarge_muon_2017
266 prop_binCRTT_electron2018_bin2
267 prop_binCRTT_muon2018_bin10
268 prop_binCROS_muon2017_bin14
269 prop_binCRTT_muon2017_bin2
270 prop_binSR_muon2016M_bin16_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

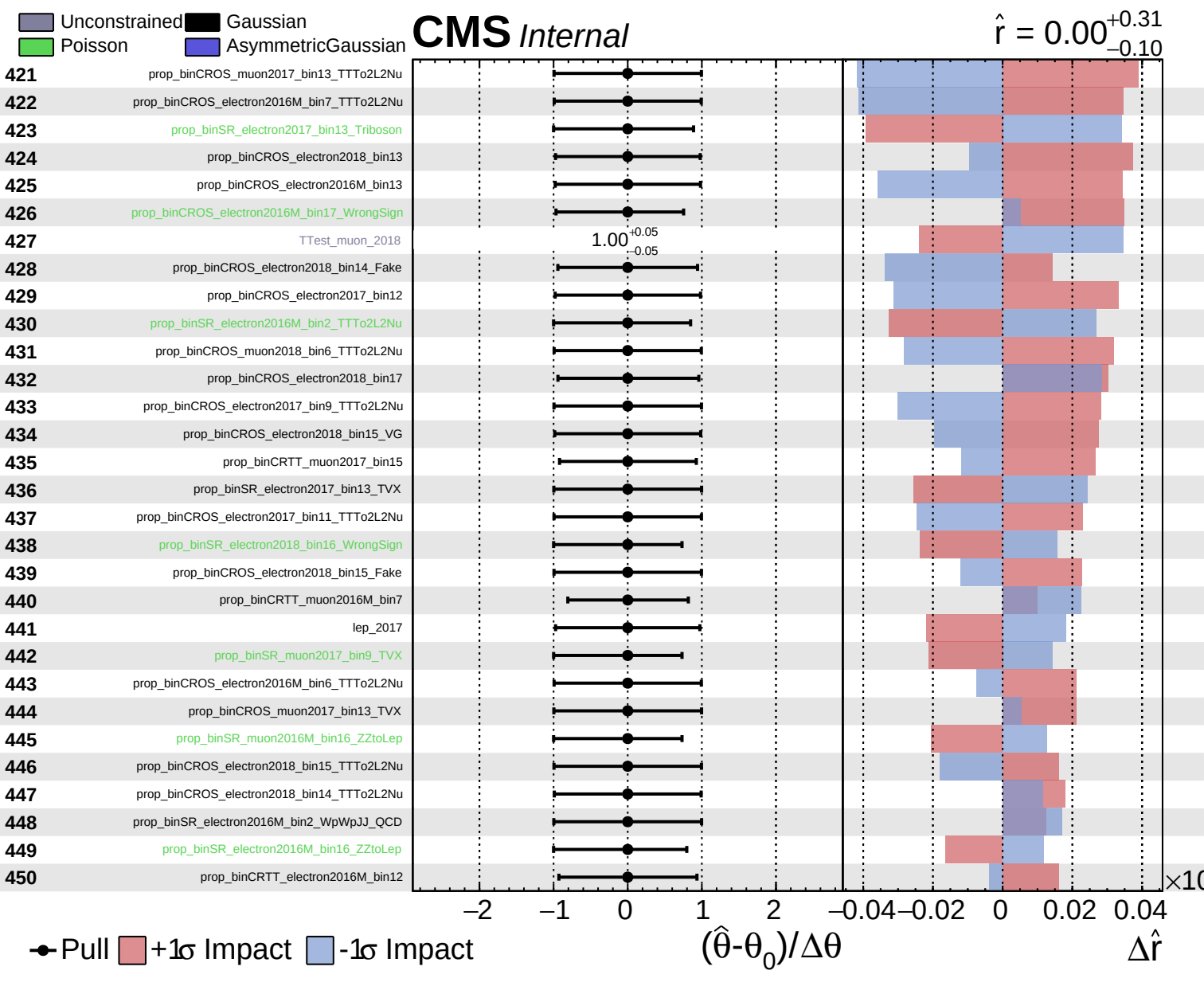

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
271 tau_vsjet_2018
272 prop_binSR_electron2017_bin13_TTTo2L2Nu
273 prop_binCRTT_muon2017_bin17
274 prop_binCRTT_electron2018_bin16
275 prop_binSR_muon2016M_bin16_TVX
276 prop_binCRTT_electron2017_bin11
277 prop_binCRTT_muon2017_bin13
278 prop_binCROS_electron2016M_bin14
279 prop_binCRTT_muon2018_bin7
280 prop_binCROS_electron2018_bin9
281 prop_binCRTT_electron2016M_bin2
282 prop_binCRTT_muon2016M_bin3
283 prop_binCROS_muon2018_bin13
284 prop_binCROS_electron2018_bin10_Fake
285 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
286 prop_binSR_electron2017_bin13_ZZtoLep
287 prop_binCRTT_electron2016M_bin8
288 prop_binCROS_muon2016M_bin9_Fake
289 prop_binCROS_electron2016M_bin0
290 prop_binCROS_muon2017_bin13_WrongSign
291 prop_binCRTT_electron2018_bin14
292 prop_binCROS_electron2017_bin13
293 prop_binSR_electron2018_bin0
294 prop_binCRTT_muon2016M_bin13
295 prop_binCRTT_electron2017_bin6
296 prop_binCRTT_muon2017_bin11
297 prop_binCRTT_muon2018_bin14
298 prop_binCRTT_muon2017_bin7
299 prop_binCRTT_electron2016M_bin5
300 prop_binCRTT_electron2016M_bin1
×10
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

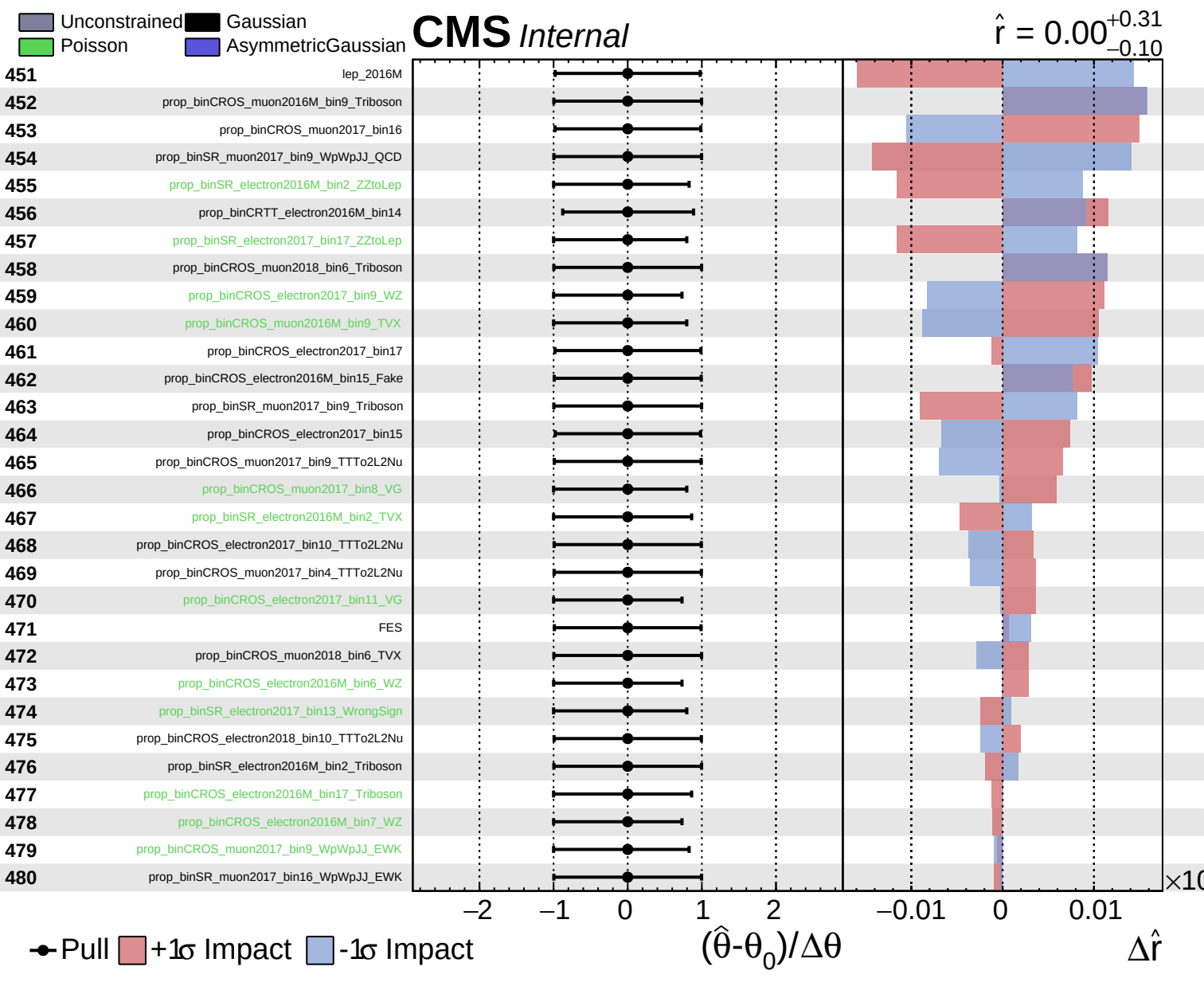

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
301 TTest_electron_2017
1.00 +0.05 −0.05
302 prop_binSR_electron2016M_bin16_TVX
303 prop_binCROS_electron2017_bin7
304 prop_binCROS_electron2016M_bin7_Fake
305 prop_binCROS_electron2018_bin3
306 prop_binCRTT_muon2018_bin5
307 prop_binCROS_muon2017_bin11
308 prop_binCRTT_muon2017_bin16
309 prop_binCRTT_electron2018_bin1
310 prop_binCRTT_electron2017_bin2
311 prop_binCROS_electron2016M_bin9_Fake
312 prop_binCRTT_muon2017_bin4
313 prop_binCRTT_electron2016M_bin6
314 prop_binCROS_muon2018_bin14
315 prop_binCRTT_muon2016M_bin4
316 prop_binCRTT_electron2018_bin11
317 prop_binCRTT_electron2018_bin0
318 prop_binCRTT_electron2017_bin8
319 prop_binCRTT_muon2016M_bin6
320 prop_binCRTT_muon2018_bin15
321 puID
322 prop_binCRTT_muon2018_bin11
323 prop_binCRTT_electron2016M_bin11
324 prop_binCRTT_electron2018_bin13
325 prop_binCRTT_electron2018_bin4
326 prop_binCROS_muon2017_bin4_Fake
327 prop_binSR_electron2018_bin16_ZZtoLep
328 prop_binCRTT_electron2018_bin7
329 prop_binCRTT_muon2018_bin12
330 prop_binCRTT_muon2017_bin12
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

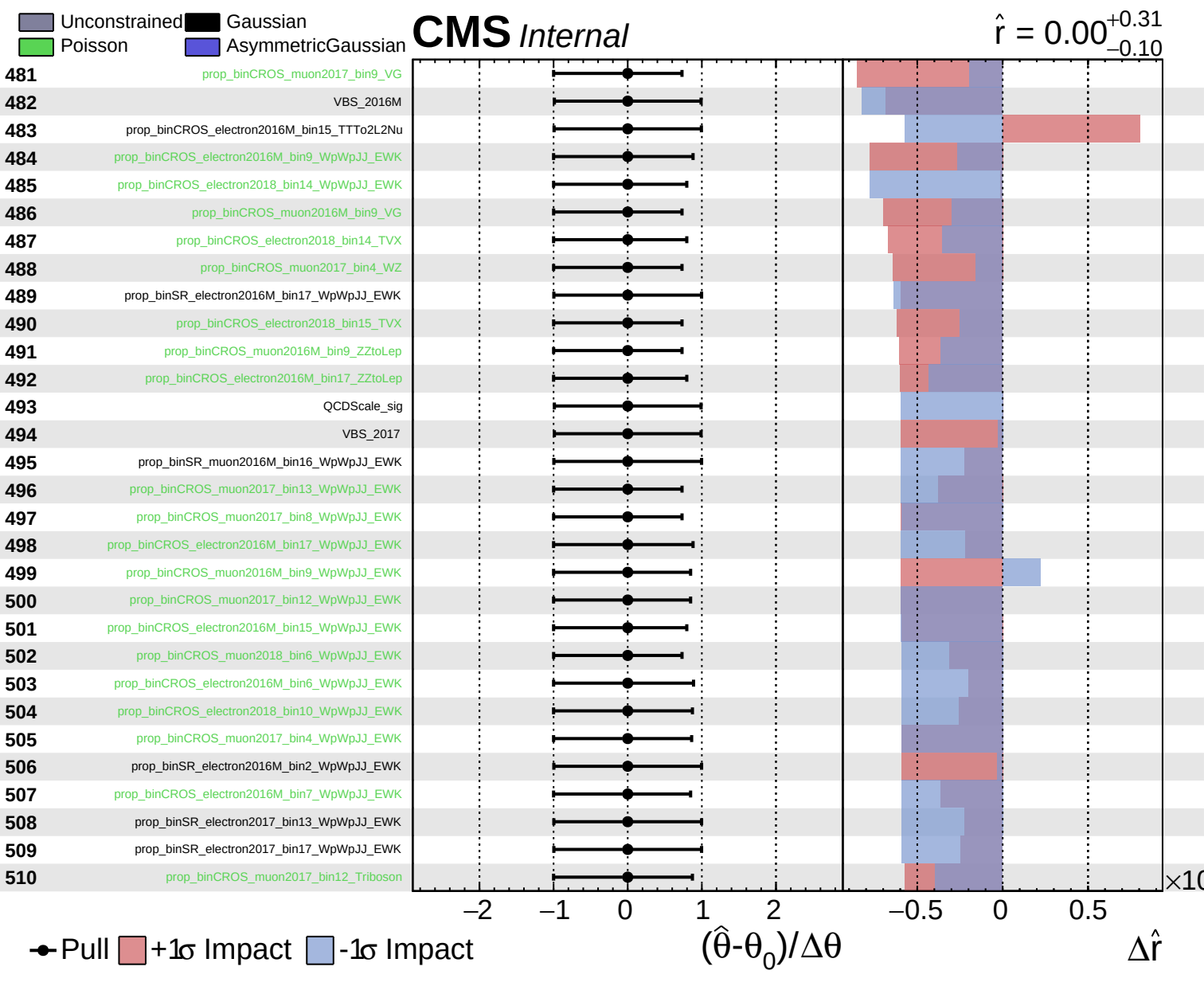

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
331 prop_binCRTT_electron2016M_bin4
332 prop_binCROS_muon2017_bin8_Fake
333 prop_binCROS_muon2017_bin13_Fake
334 prop_binCROS_electron2017_bin11_WrongSign
335 mischarge_muon_2018
336 prop_binCROS_muon2017_bin4_WrongSign
337 prop_binCRTT_electron2017_bin3
338 prop_binCRTT_muon2018_bin6
339 prop_binSR_electron2016M_bin16_WrongSign
340 prop_binCRTT_electron2017_bin4
341 prop_binCROS_muon2018_bin0
342 prop_binCRTT_electron2018_bin8
343 prop_binSR_electron2016M_bin16_Triboson
344 prop_binCROS_electron2016M_bin8
345 prop_binCRTT_electron2017_bin14
346 prop_binCRTT_muon2016M_bin15
347 prop_binCRTT_electron2017_bin12
348 mischarge_electron_2016M
349 prop_binCRTT_electron2016M_bin15
350 prop_binCROS_muon2017_bin15
351 prop_binCROS_muon2018_bin15
352 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
353 prop_binCRTT_muon2018_bin17
354 TTest_muon_2016M
1.00 +0.06 −0.06
355 prop_binCROS_muon2017_bin7
356 prop_binCRTT_electron2018_bin6
357 prop_binCROS_electron2017_bin10_WrongSign
358 TTest_muon_2017
1.00 +0.05 −0.05
359 prop_binCROS_muon2017_bin17
360 prop_binCRTT_electron2016M_bin9
×10
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

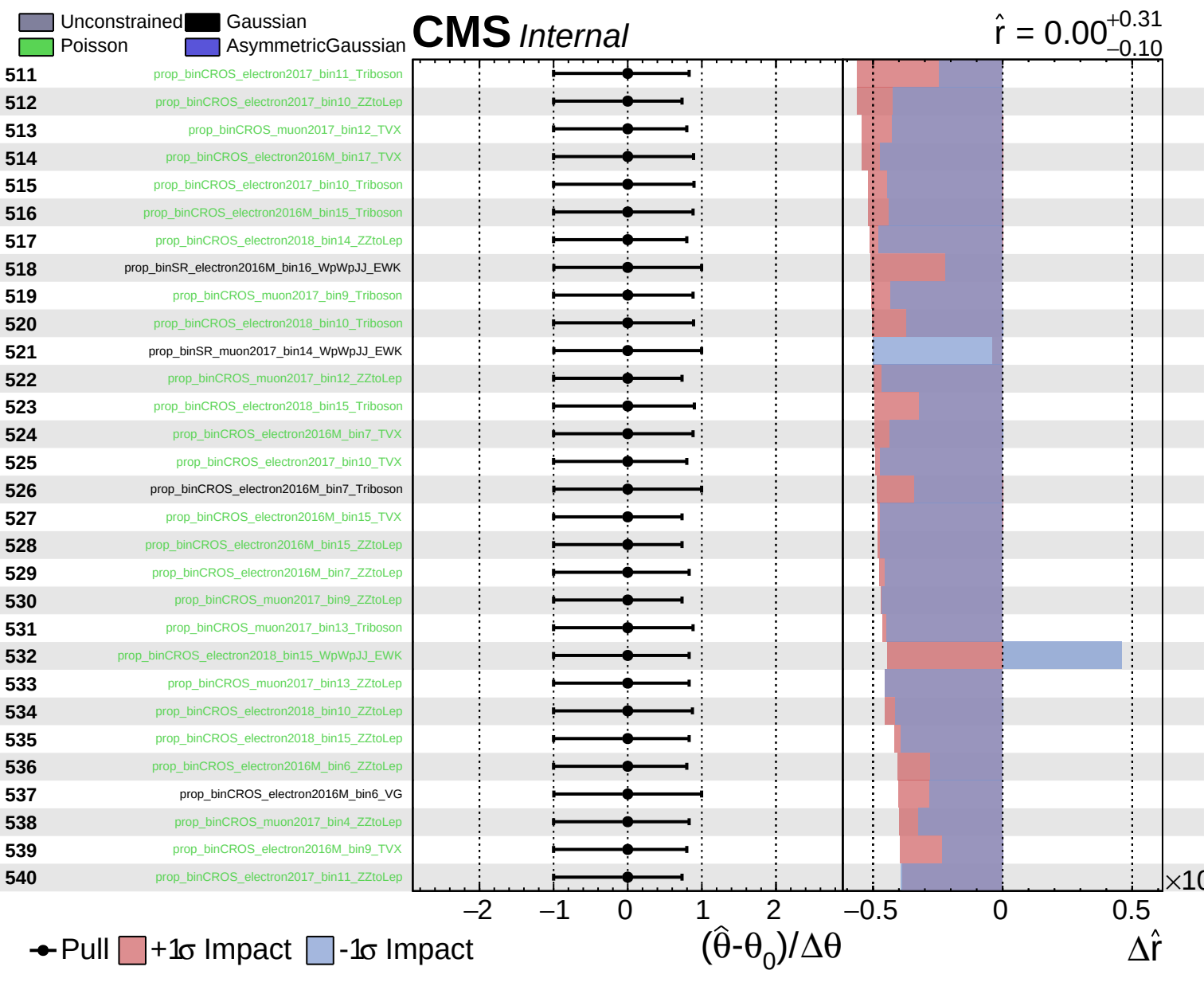

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
361 prop_binCRTT_muon2017_bin9
362 QCDScale_ZZtoLep
363 prop_binCRTT_muon2018_bin16
364 prop_binCRTT_electron2016M_bin17
365 prop_binCROS_electron2018_bin10_WrongSign
366 prop_binCRF_electron2016M_bin0
367 prop_binCROS_muon2017_bin12_Fake
368 prop_binCROS_electron2016M_bin6_Fake
369 prop_binCRTT_muon2016M_bin16
370 prop_binCROS_muon2017_bin9_Fake
371 prop_binCROS_muon2017_bin9_WrongSign
372 prop_binCROS_electron2017_bin10_Fake
373 prop_binSR_electron2016M_bin17_ZZtoLep
374 prop_binCRTT_electron2017_bin17
375 prop_binSR_electron2017_bin17_WpWpJJ_QCD
376 prop_binCRTT_muon2017_bin0
377 TTest_electron_2018
1.00 +0.05 −0.05
378 prop_binCROS_electron2016M_bin16
379 TTest_electron_2016M
1.00 +0.06 −0.06
380 prop_binCRTT_electron2016M_bin13
381 prop_binCROS_muon2016M_bin16
382 prop_binSR_muon2017_bin9_TTTo2L2Nu
383 prop_binCROS_electron2018_bin14_WrongSign
384 prop_binCRTT_muon2016M_bin14
385 prop_binCRTT_muon2018_bin9
386 prop_binCROS_electron2016M_bin10
387 prop_binCROS_electron2018_bin16
388 prop_binCROS_muon2016M_bin15
389 prop_binCRTT_muon2017_bin14
390 prop_binCRTT_muon2016M_bin8
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

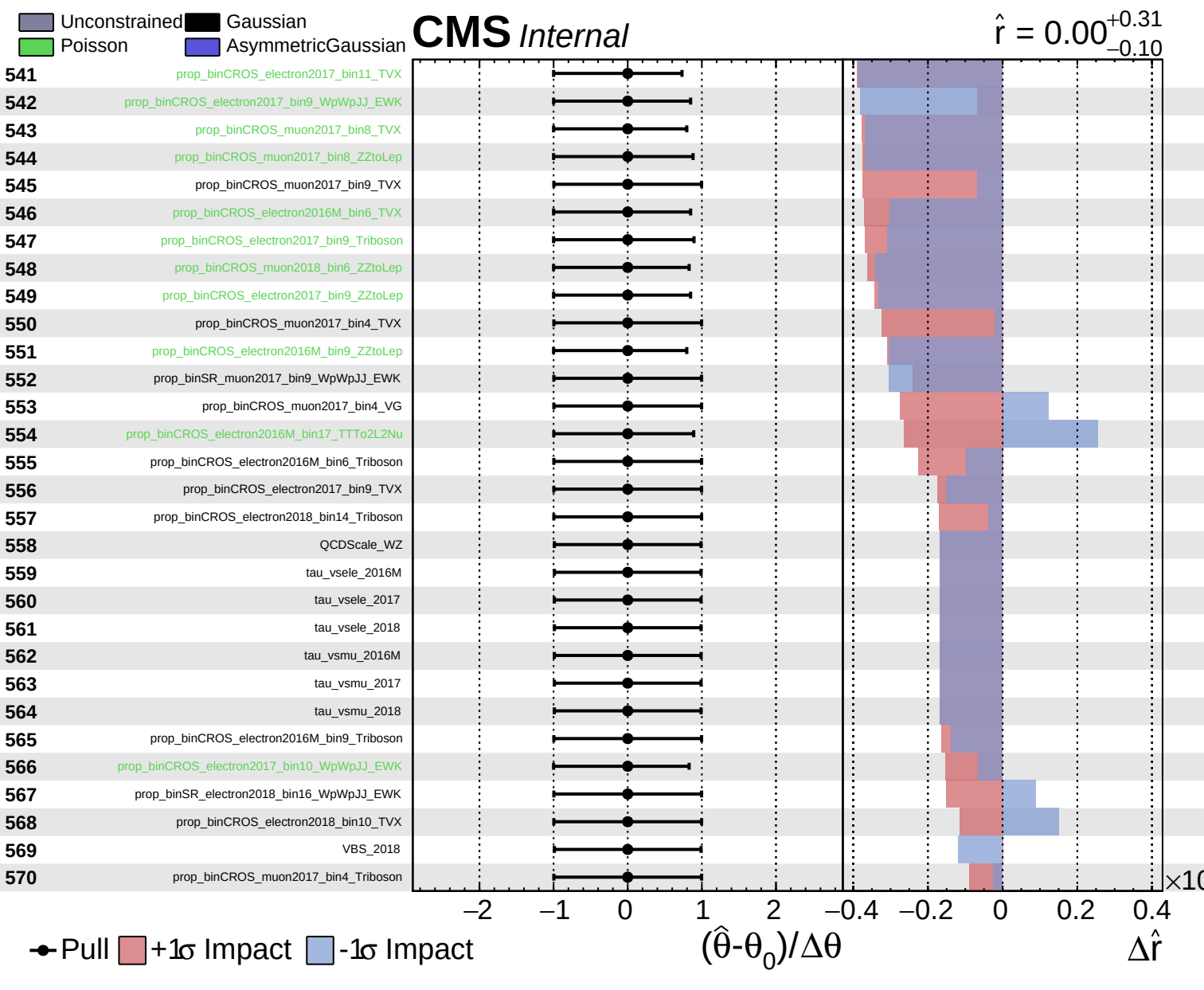

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00+0.31−0.10
391 prop_binCRTT_muon2016M_bin9
392 mischarge_electron_2017
393 prop_binCROS_muon2016M_bin17
394 prop_binCRTT_muon2016M_bin12
395 prop_binCROS_muon2017_bin12_TTTo2L2Nu
396 prop_binCROS_electron2017_bin16
397 prop_binCROS_electron2016M_bin11
398 prop_binCROS_electron2017_bin11_Fake
399 prop_binSR_electron2017_bin17_TVX
400 prop_binCROS_muon2016M_bin14
401 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
402 prop_binCRTT_muon2017_bin10
403 prop_binCRTT_electron2017_bin16
404 prop_binCROS_electron2017_bin14
405 prop_binCROS_muon2016M_bin12
406 mischarge_electron_2018
407 prop_binSR_electron2017_bin13_WpWpJJ_QCD
408 prop_binCROS_muon2018_bin17
409 prop_binCRTT_muon2016M_bin17
410 prop_binCROS_muon2018_bin16
411 prop_binCRTT_electron2018_bin9
412 prop_binCRTT_muon2016M_bin10
413 lep_2018
414 prop_binCRTT_electron2017_bin15
415 prop_binSR_electron2016M_bin2_WZ
416 prop_binCROS_muon2017_bin8_TTTo2L2Nu
417 prop_binCRTT_muon2017_bin8
418 prop_binCROS_muon2016M_bin13
419 prop_binCRTT_electron2016M_bin16
420 prop_binSR_electron2016M_bin17_WrongSign
×10
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

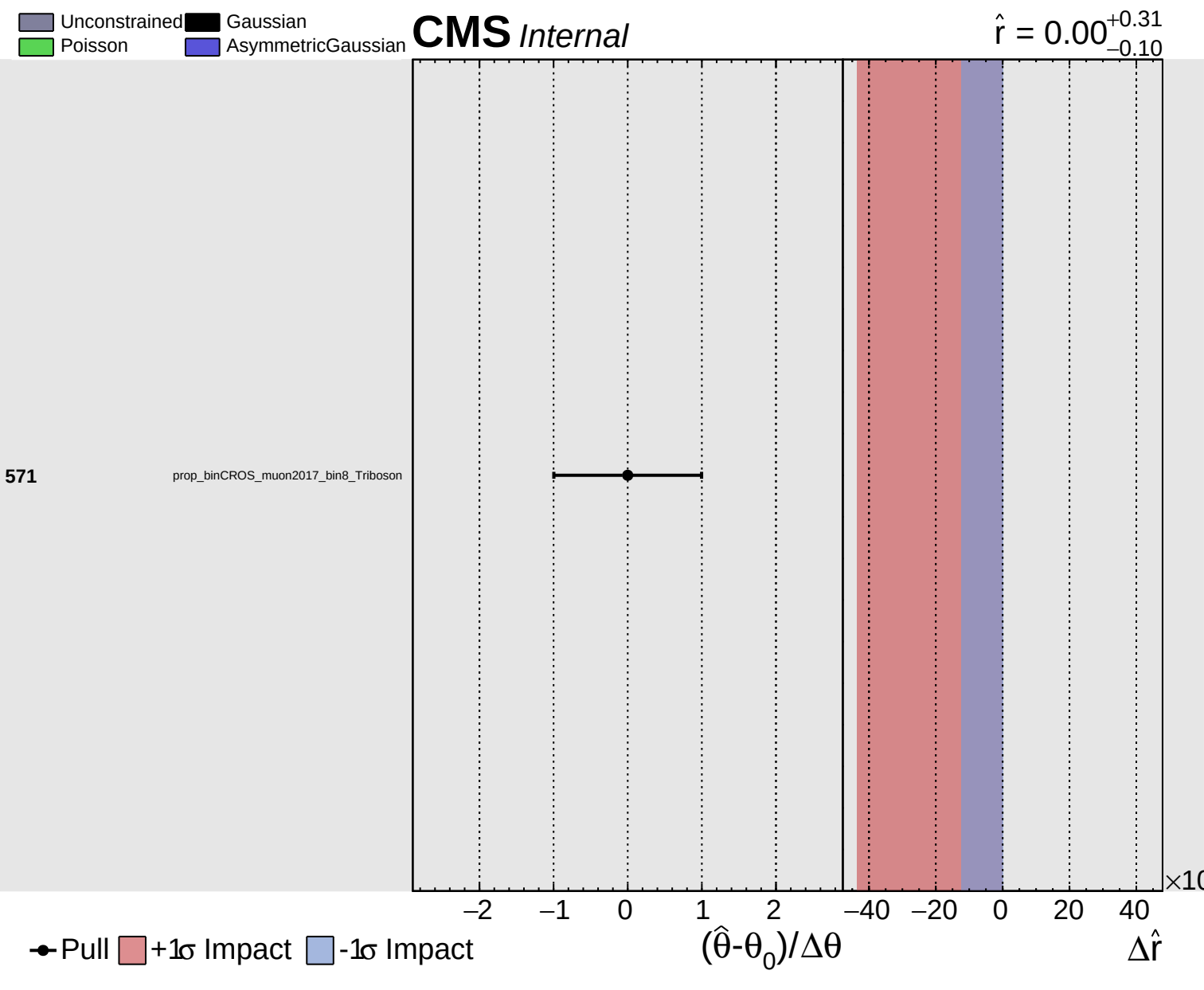

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
421 prop_binCROS_muon2017_bin13_TTTo2L2Nu
422 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
423 prop_binSR_electron2017_bin13_Triboson
424 prop_binCROS_electron2018_bin13
425 prop_binCROS_electron2016M_bin13
426 prop_binCROS_electron2016M_bin17_WrongSign
427 TTest_muon_2018
1.00 +0.05 −0.05
428 prop_binCROS_electron2018_bin14_Fake
429 prop_binCROS_electron2017_bin12
430 prop_binSR_electron2016M_bin2_TTTo2L2Nu
431 prop_binCROS_muon2018_bin6_TTTo2L2Nu
432 prop_binCROS_electron2018_bin17
433 prop_binCROS_electron2017_bin9_TTTo2L2Nu
434 prop_binCROS_electron2018_bin15_VG
435 prop_binCRTT_muon2017_bin15
436 prop_binSR_electron2017_bin13_TVX
437 prop_binCROS_electron2017_bin11_TTTo2L2Nu
438 prop_binSR_electron2018_bin16_WrongSign
439 prop_binCROS_electron2018_bin15_Fake
440 prop_binCRTT_muon2016M_bin7
441 lep_2017
442 prop_binSR_muon2017_bin9_TVX
443 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
444 prop_binCROS_muon2017_bin13_TVX
445 prop_binSR_muon2016M_bin16_ZZtoLep
446 prop_binCROS_electron2018_bin15_TTTo2L2Nu
447 prop_binCROS_electron2018_bin14_TTTo2L2Nu
448 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
449 prop_binSR_electron2016M_bin16_ZZtoLep
450 prop_binCRTT_electron2016M_bin12
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
451 lep_2016M
452 prop_binCROS_muon2016M_bin9_Triboson
453 prop_binCROS_muon2017_bin16
454 prop_binSR_muon2017_bin9_WpWpJJ_QCD
455 prop_binSR_electron2016M_bin2_ZZtoLep
456 prop_binCRTT_electron2016M_bin14
457 prop_binSR_electron2017_bin17_ZZtoLep
458 prop_binCROS_muon2018_bin6_Triboson
459 prop_binCROS_electron2017_bin9_WZ
460 prop_binCROS_muon2016M_bin9_TVX
461 prop_binCROS_electron2017_bin17
462 prop_binCROS_electron2016M_bin15_Fake
463 prop_binSR_muon2017_bin9_Triboson
464 prop_binCROS_electron2017_bin15
465 prop_binCROS_muon2017_bin9_TTTo2L2Nu
466 prop_binCROS_muon2017_bin8_VG
467 prop_binSR_electron2016M_bin2_TVX
468 prop_binCROS_electron2017_bin10_TTTo2L2Nu
469 prop_binCROS_muon2017_bin4_TTTo2L2Nu
470 prop_binCROS_electron2017_bin11_VG
471 FES
472 prop_binCROS_muon2018_bin6_TVX
473 prop_binCROS_electron2016M_bin6_WZ
474 prop_binSR_electron2017_bin13_WrongSign
475 prop_binCROS_electron2018_bin10_TTTo2L2Nu
476 prop_binSR_electron2016M_bin2_Triboson
477 prop_binCROS_electron2016M_bin17_Triboson
478 prop_binCROS_electron2016M_bin7_WZ
479 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
480 prop_binSR_muon2017_bin16_WpWpJJ_EWK
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
481 prop_binCROS_muon2017_bin9_VG
482 VBS_2016M
483 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
484 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
485 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
486 prop_binCROS_muon2016M_bin9_VG
487 prop_binCROS_electron2018_bin14_TVX
488 prop_binCROS_muon2017_bin4_WZ
489 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
490 prop_binCROS_electron2018_bin15_TVX
491 prop_binCROS_muon2016M_bin9_ZZtoLep
492 prop_binCROS_electron2016M_bin17_ZZtoLep
493 QCDScale_sig
494 VBS_2017
495 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
496 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
497 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
498 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
499 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
500 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
501 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
502 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
503 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
504 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
505 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
506 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
507 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
508 prop_binSR_electron2017_bin13_WpWpJJ_EWK
509 prop_binSR_electron2017_bin17_WpWpJJ_EWK
510 prop_binCROS_muon2017_bin12_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
511 prop_binCROS_electron2017_bin11_Triboson
512 prop_binCROS_electron2017_bin10_ZZtoLep
513 prop_binCROS_muon2017_bin12_TVX
514 prop_binCROS_electron2016M_bin17_TVX
515 prop_binCROS_electron2017_bin10_Triboson
516 prop_binCROS_electron2016M_bin15_Triboson
517 prop_binCROS_electron2018_bin14_ZZtoLep
518 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
519 prop_binCROS_muon2017_bin9_Triboson
520 prop_binCROS_electron2018_bin10_Triboson
521 prop_binSR_muon2017_bin14_WpWpJJ_EWK
522 prop_binCROS_muon2017_bin12_ZZtoLep
523 prop_binCROS_electron2018_bin15_Triboson
524 prop_binCROS_electron2016M_bin7_TVX
525 prop_binCROS_electron2017_bin10_TVX
526 prop_binCROS_electron2016M_bin7_Triboson
527 prop_binCROS_electron2016M_bin15_TVX
528 prop_binCROS_electron2016M_bin15_ZZtoLep
529 prop_binCROS_electron2016M_bin7_ZZtoLep
530 prop_binCROS_muon2017_bin9_ZZtoLep
531 prop_binCROS_muon2017_bin13_Triboson
532 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
533 prop_binCROS_muon2017_bin13_ZZtoLep
534 prop_binCROS_electron2018_bin10_ZZtoLep
535 prop_binCROS_electron2018_bin15_ZZtoLep
536 prop_binCROS_electron2016M_bin6_ZZtoLep
537 prop_binCROS_electron2016M_bin6_VG
538 prop_binCROS_muon2017_bin4_ZZtoLep
539 prop_binCROS_electron2016M_bin9_TVX
540 prop_binCROS_electron2017_bin11_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
541 prop_binCROS_electron2017_bin11_TVX
542 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
543 prop_binCROS_muon2017_bin8_TVX
544 prop_binCROS_muon2017_bin8_ZZtoLep
545 prop_binCROS_muon2017_bin9_TVX
546 prop_binCROS_electron2016M_bin6_TVX
547 prop_binCROS_electron2017_bin9_Triboson
548 prop_binCROS_muon2018_bin6_ZZtoLep
549 prop_binCROS_electron2017_bin9_ZZtoLep
550 prop_binCROS_muon2017_bin4_TVX
551 prop_binCROS_electron2016M_bin9_ZZtoLep
552 prop_binSR_muon2017_bin9_WpWpJJ_EWK
553 prop_binCROS_muon2017_bin4_VG
554 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
555 prop_binCROS_electron2016M_bin6_Triboson
556 prop_binCROS_electron2017_bin9_TVX
557 prop_binCROS_electron2018_bin14_Triboson
558 QCDScale_WZ
559 tau_vsele_2016M
560 tau_vsele_2017
561 tau_vsele_2018
562 tau_vsmu_2016M
563 tau_vsmu_2017
564 tau_vsmu_2018
565 prop_binCROS_electron2016M_bin9_Triboson
566 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
567 prop_binSR_electron2018_bin16_WpWpJJ_EWK
568 prop_binCROS_electron2018_bin10_TVX
569 VBS_2018
570 prop_binCROS_muon2017_bin4_Triboson
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.31 −0.10
571
prop_binCROS_muon2017_bin8_Triboson
×10
−2
−1
0
1
2
−40
−20
0
20
40
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂